

From Legal PDFs to Verifiable Answers

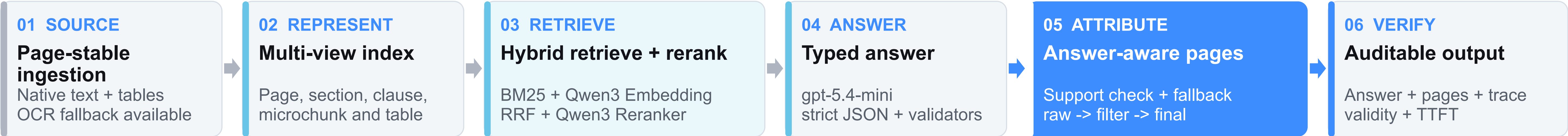
Evaluating answer quality, page-level grounding and latency across 100 legal questions

HYPOTHESIS
Can a page-traceable Legal RAG pipeline balance answer quality, source grounding, validity and latency?
H2: Does answer-aware attribution sharpen citations while retaining at least one correct page per annotated question?

FINDING
H1 is partly supported: grounding missed its 0.85 threshold. H2 holds for precision, compactness and 95/95 hit coverage.

METHODOLOGY

100 questions | 30 publicly available legal PDFs | 590 pages | 95 with page annotations



CONTINUOUS PROVENANCE | document_id + page_number

The same identifiers survive every stage, so a final citation can be traced back to the exact PDF page.

Evaluation A: end-to-end profile on 100 questions. Evaluation B: paired raw -> filter -> final page sets on the same 95 annotated questions.

RESULTS | SYSTEM
Grounding limits an otherwise strong system

0.956

0.847

1.000

0.938

=

0.759

answer quality

grounding
n = 100

valid outputs

latency multiplier

total score

0.847 < 0.850

100/100 valid outputs

Mean TTFT
6.39 s

0.943 deterministic | 0.987 free text

H1 is partly supported: quality, validity and latency pass; benchmark grounding misses 0.85.

RESULTS | ATTRIBUTION
Answer-aware selection removes citation noise

Macro averages | n = 95 annotated questions

	RAW	FILTER	FINAL
Precision	63.8%	63.8%	73.9%
Recall	96.8%	96.2%	95.2%
F-beta	87.7%	87.3%	89.2%
Pages / Q	2.56	2.54	2.23

+10.2 pp
precision
95% CI +6.3 to +14.4

-0.33
pages / question
95% CI -0.46 to -0.20

95/95
hit coverage retained
F-beta gain uncertain

RESULTS | BOUNDARY
Narrowing helps local - and hurts distributed evidence

Exploratory F-beta change, raw -> final

Single page n = 59	+4.0 pp
Multi page n = 36	-2.6 pp
Single document n = 60	+3.2 pp
Multi document n = 35	-1.4 pp

-3 pp0+5 pp

Multi-page: -2.6 pp - the clearest failure boundary.

Do not conflate: benchmark grounding 0.847 = all 100 answers; macro page F-beta 0.892 = final attribution on 95 annotated questions.

DECISION

Retrieve broadly. Attribute explicitly. Narrow conservatively when evidence spans pages or documents.

LIMITATIONS

ONE SAVED RUN
Retrospective thresholds and questions; no repeated seeds.

NO CAUSAL ABLATION
Pipeline stages were not independently compared against a baseline.

LIMITED GENERALITY
TTFT only; OCR not exercised; one DIFC corpus and one language setting.